Piyush Bhuyan

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Summary

Applied AI Engineer with 1.5+ years building production AI systems, machine learning pipelines, and scalable backend services. Focused on multi-agent systems, RAG, observability, and cloud-native architectures.

Work Experience

Keyloop India | Hyderabad

Applied AI Engineer Jul 2026 – Present

- Architecting an autonomous incident orchestration platform spanning **38 enterprise products** with a **6-tier deterministic-AI cascade**, integrating Claude function calling, multi-agent orchestration, and BM25 pattern matching, routing **80% of events at $0 LLM cost** via deterministic rules, targeting **75%+ zero-touch autonomous resolution** at a blended **$0.03/event AI cost**.
- Reduced P2 MTTR by **99%** and major-incident resolution time by **97%** by correlating real-time signals, change activity, and cross-team case data through parallel enrichment, with autonomous remediation via Paramiko-based execution agents.
- Advancing **AI maturity from S0 (fully manual) to S3 (self-improving)** via a weekly pipeline using KMeans clustering, Titan, and Sonnet, compressing incident-pattern-to-runbook cycles from **weeks to under 7 days** with cosine-similarity quality gates.
- Guiding **4 engineers across HR, Apps, and Infrastructure** on AI engineering best practices, GDPR/PII safeguards, and LLM workflow optimization, achieving a **93% reduction in token consumption** through prompt and workflow improvements.

Keyloop India | Hyderabad

Associate Engineer, AI Enablement Jul 2025 – Jun 2026

- Shipped a full-stack AI operations platform (FastAPI + React/TypeScript + MySQL + Docker) with **40+ REST API endpoints** and Grafana/Prometheus observability, achieving **40x throughput improvement** (5 to 200 req/s) via thread-safe caching, MySQL connection pooling (48 connections), and GZIP compression reducing payload size **90%**, scaling to **500+ concurrent engineers**.
- Eliminated an N+1 query bottleneck by replacing 31 sequential database calls with batch `IN`-clause queries and a thread-safe connection pool, reducing critical-path latency **620x** and enabling **50x user capacity growth** (10 to 500 concurrent users).
- Built a Python/FastAPI microservice for automated IP/hostname resolution using a 4-stage ML extraction pipeline, LDAP-backed JWT authentication, and RBAC authorization, deployed via GitHub Actions CI/CD, saving **1,000+ engineering hours annually**.

Skills

Languages & Backend: Python, FastAPI, REST APIs, MySQL, MongoDB, Git

LLM & Agentic AI: Claude, Bedrock, Prompt Engineering, RAG, Multi-agent Systems, LangChain, LangGraph, ChromaDB, Pinecone

Cloud & Infrastructure: AWS (Lambda, DynamoDB, SQS, S3, Bedrock), Docker, Kubernetes, Nginx, GitHub Actions, CI/CD

Machine Learning & MLOps: Scikit-learn, XGBoost, PyTorch, Transformers, Fine-tuning, NLP, Pandas, NumPy, MLflow, DVC, Evals

Core Competencies: System Design, Distributed Systems, Agentic Workflows, Agile, Data Structures & Algorithms

Projects

pispyre | *Python, Static Analysis, AST, CLI Development, SQLite, Ruff, MyPy, Pytest* | [Repo] 2026

- Built **pispyre**, an open-source static analysis CLI that detects hallucinated imports in AI-generated Python code before runtime, deployable as a **CI gate, Python library, and pre-commit hook**.
- Engineered a modular rule engine implementing six analyses for phantom packages, removed stdlib modules, invalid attributes, abandoned dependencies, and moved symbols, with JSON and SARIF reporting for automated code quality workflows.

Layered Security - Gradient Boosting Meets Naive Bayes for Intrusion Detection | Springer CIS 2025 | [Link] 2026

- Developed a double-layered intrusion detection system combining **Naive Bayes, XGBoost, and GAN-based augmentation**, improving minority-class detection by **14.49pp for R2L** and **5.42pp for U2R** on NSL-KDD.
- Replaced SVM with **XGBoost** and augmented rare attack classes using GANs, achieving an **80.20% F1-score** with reported gains of **4.5% in recall** and **5.4% in false-alarm rate** over the SVM-based second layer.

Education

Vasavi College of Engineering, Hyderabad 2021 – 2025

- *Bachelor of Engineering, Information Technology* **CGPA:** 9.04/10

Certifications

Microsoft Certified: Machine Learning Operations Engineer Associate (AI-300) | Microsoft 2026

AWS Certified Cloud Practitioner (CLF-C02) | AWS 2026